Neuromorphic Computing project proposal

Group 2

Marije Bekema s1106132
Karan Chand s1033357
Freek Hens s1033509
Suzan Lejeune s1015354

Radboud University, Nijmegen, The Netherlands

The aim of the project is to replicate the spoken digit classification SNN described by Dong et al. in Nengo.

This project is inspired by earlier done research by Dong et al., which showed high performance in two different situations (97.5% and 93.8% resp.).

The SNN will be implemented in Nengo. If possible, the TIDIGITS dataset as used by Dong et al. will be used, however, if this proves infeasible, it is still interesting to see if the model achieves the same results on different datasets. Possible other audio classification datasets can be found on:
https://huggingface.co/datasets?task_categories=task_categories:audio-classification&sort=downloads

References

Dong M, Huang X, Xu B (2018) Unsupervised speech recognition through spike-timing-dependent plasticity in a convolutional spiking neural network. PLoS ONE 13(11): e0204596. https://doi.org/10.1371/journal.pone.0204596